

PRACTICAL-02

```
#include <iostream>
```

```
#include <vector>
```

```
#include <chrono>
```

```
using namespace std;
```

```
using namespace std::chrono;
```

```
// Linear Search Function
```

```
int linearSearch(const vector<int>& arr, int key)
```

```
{
```

```
    for (int i = 0; i < arr.size(); i++)
```

```
    {
```

```
        if (arr[i] == key)
```

```
            return i;
```

```
    }
```

```
    return -1;
```

```
}
```

```
// Binary Search Function
```

```
int binarySearch(const vector<int>& arr, int key)
```

```
{
```

```
    int low = 0;
```

```
    int high = arr.size() - 1;
```

PRACTICAL-02

```
while (low <= high)

{

    int mid = low + (high - low) / 2;


    if (arr[mid] == key)

        return mid;

    else if (arr[mid] < key)

        low = mid + 1;

    else

        high = mid - 1;

}


return -1;

}


int main()

{

    int n;

    cout << "Enter the size of the array: ";

    cin >> n;


    vector<int> arr(n);
```

PRACTICAL-02

```
for (int i = 0; i < n; i++)
```

```
    arr[i] = i + 1;
```

```
int key;
```

```
cout << "Enter the element to search: ";
```

```
cin >> key;
```

```
if (key < 1 || key > n)
```

```
{
```

```
    cout << "\nElement is outside the array range.\n";
```

```
    return 0;
```

```
}
```

```
int index;
```

```
// Linear Search
```

```
auto start = high_resolution_clock::now();
```

```
index = linearSearch(arr, key);
```

```
auto stop = high_resolution_clock::now();
```

```
auto linearTime = duration_cast<microseconds>(stop - start);
```

PRACTICAL-02

```
cout << "\n==== Linear Search =====\n";

if (index != -1)

    cout << "Element found at index: " << index << endl;

else

    cout << "Element not found." << endl;


cout << "Execution Time: " << linearTime.count()

    << " microseconds\n";


// Binary Search

start = high_resolution_clock::now();

index = binarySearch(arr, key);

stop = high_resolution_clock::now();


auto binaryTime = duration_cast<microseconds>(stop - start);


cout << "\n==== Binary Search =====\n";

if (index != -1)

    cout << "Element found at index: " << index << endl;

else

    cout << "Element not found." << endl;


cout << "Execution Time: " << binaryTime.count()
```

PRACTICAL-02

```
<< " microseconds\n";

// Comparison

cout << "\n===== Performance Comparison =====\n";

if (linearTime > binaryTime)

    cout << "Binary Search is faster than Linear Search.\n";

else if (linearTime < binaryTime)

    cout << "Linear Search is faster than Binary Search.\n";

else

    cout << "Both searches took the same time.\n";

return 0;

}
```